Nano Vibration Energy Harvester: Harvesting Tiny Vibrations into Usable Energy for Smart and Sustainable Spaces

1st Tanjina Akter Nisa
Dept. of Computer Science and Engineering
University of Asia Pacific
Dhaka, Bangladesh
22201086@uap-bd.edu

2nd Md. Daudul Islam
Dept. of Computer Science and Engineering
University of Asia Pacific
Dhaka, Bangladesh
22201088@uap-bd.edu

3rd Saswata Ghosal
Dept. of Computer Science and Engineering
University of Asia Pacific
Dhaka, Bangladesh
22201090@uap-bd.edu

***Abstract*—The rapid proliferation of portable electronic devices has increased the demand for frequent charging, leading to greater reliance on grid electricity and consequently higher carbon emissions. This project presents a Nano Vibration Energy Harvester that captures ambient mechanical vibrations using piezoelectric sensors and converts them into usable electrical energy. The harvested AC voltage is rectified to DC through a bridge rectifier, stabilized via capacitors and a 2S charging board to prevent overvoltage, and stored in a 7.2V Li-ion battery. The stored energy is utilized to power multiple applications, including a mini power bank for 5V DC mobile charging, a 12V to 220V inverter for AC loads, and an LM2596 buck converter to supply regulated 5V DC to an ESP32 microcontroller for IoT-based monitoring and real-time visualization. Experimental results indicate that after five simulated footsteps, the system produced a maximum output of 0.271 V, peak voltage of 2.976 V, and a total harvested energy of 0.023 J over a duration of 70 seconds. This work demonstrates the practical feasibility of converting ambient kinetic energy into usable power for low-energy devices while promoting sustainable energy utilization and showcasing embedded system design with microcontroller interfacing.**

***Index Terms*—Piezoelectric Sensors, Energy Harvesting, ESP32 Microcontroller, IoT Monitoring, Sustainable Energy, Embedded Systems**

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I. INTRODUCTION

A. Background and Motivation

The rapid proliferation of portable electronic devices, including smartphones, wearable gadgets, and Internet-of-Things (IoT) sensors, has led to an unprecedented demand for energy. These devices predominantly rely on conventional grid electricity for operation and charging. However, traditional energy generation methods, such as fossil-fuel-based power plants, contribute significantly to environmental degradation, greenhouse gas emissions, and global climate change [1]. Consequently, there is an increasing emphasis on identifying alternative, sustainable energy sources that can mitigate the environmental impact of modern electronics. Among various renewable energy strategies, ambient mechanical vibrations—originating from human activities such as footsteps, vehicular movements, industrial machinery, and structural oscillations—represent a largely untapped and continuous source of energy. The conversion of mechanical vibrations into usable electrical energy presents an opportunity to power low-energy devices while simultaneously promoting green and sustainable technologies [2], [3]. Piezoelectric materials, due to their inherent electromechanical coupling properties, provide a promising solution for harvesting ambient mechanical energy. When subjected to mechanical stress, these materials generate an electrical charge, which can then be conditioned and stored for practical use. Recent studies have demonstrated the feasibility of floor-integrated piezoelectric tiles, cantilever structures, and sensor arrays designed specifically to harvest energy from footsteps and other environmental vibrations [**?**], [4], [5]. By leveraging such energy-harvesting techniques, it becomes possible to develop self-sustaining systems that reduce dependency on the electrical grid, extend device lifetimes, and contribute to the advancement of smart urban infrastructure.

B. Problem Statement

Despite significant research in vibration energy harvesting, current solutions often face challenges related to low conversion efficiency, limited scalability, and inconsistent power output. Small-scale implementations, such as individual piezoelectric units, frequently fail to generate sufficient energy to reliably power IoT devices or portable electronics. Furthermore, many systems lack effective energy management strategies, including AC-to-DC conversion, voltage regulation, and energy storage, thereby limiting their practical applicability [**?**]. There is a pressing need for an integrated and scalable energy-harvesting system capable of converting ambient mechanical vibrations into stable electrical energy. Such a system should incorporate effective energy storage mechanisms, provide regulated output for electronic devices, and facilitate real-time monitoring of energy generation and consumption.

C. Objectives

The primary objectives of this study are as follows:

- To harvest electrical energy from ambient mechanical vibrations using piezoelectric transducers.
- To convert the generated alternating current (AC) into direct current (DC) and regulate it for safe and efficient storage.
- To supply harvested energy to low-power devices, IoT modules, and small AC loads, thereby demonstrating practical usability.
- To implement a real-time monitoring system using an ESP32 microcontroller, enabling visualization of energy generation, storage, and consumption metrics.
- To analyze the overall system performance, assess energy conversion efficiency, and explore strategies for scalability in larger deployments.

D. Scope of the Project

This project primarily focuses on indoor environments such as offices, gyms, airports, university corridors, and other public or semi-public spaces where mechanical vibrations are readily available. The project encompasses the following aspects:

- Design and implementation of hardware circuits incorporating piezoelectric transducers, energy conditioning circuits, storage units, and output interfaces.
- Development of firmware for the ESP32 microcontroller to facilitate real-time monitoring, data acquisition, and visualization of energy metrics.
- Evaluation of energy conversion efficiency under varying vibration intensities, frequencies, and load conditions.
- Assessment of system scalability and adaptability to larger spaces and higher device counts.
- Exploration of the potential integration of the proposed system into smart building infrastructures and sustainable urban solutions.

II. Literature Review

Energy harvesting from ambient mechanical vibrations has attracted significant attention in recent years due to its potential to provide sustainable power for low-energy electronic systems. Several studies have explored piezoelectric-based solutions for converting human footsteps and other vibration sources into usable electrical energy.

Althubiti, Alam, and Hoque [1] investigated the use of piezoelectric sensors for harvesting energy from human footsteps. Their research demonstrated that piezoelectric transducers could effectively generate measurable voltage outputs when subjected to mechanical stress, enabling the operation of low-power devices such as wireless sensors and LEDs. The study emphasized the potential of integrating such systems in pedestrian walkways and smart flooring applications to promote renewable energy usage.

Similarly, Babu, Alam, and Hoque [2] developed an energy-harvesting floor tile using piezoelectric patches. Their work focused on optimizing tile design and sensor placement to maximize energy generation efficiency. The results showed that increasing the number of sensors and improving mechanical coupling enhanced both voltage and power output, validating the practicality of piezoelectric floors in real-world environments.

An applied example of vibration energy harvesting in public infrastructure was presented by ScienceAlert [3], which reported on Japanese innovations that generate electricity from walking. These systems utilize embedded piezoelectric materials in sidewalks and floors to capture kinetic energy from footsteps, converting it into small amounts of usable electricity for powering lighting or display systems in urban settings.

Jadhav, Patil, and Patil [4] introduced an IoT-based energy monitoring system using the ESP32 microcontroller and the Blynk platform. Their design provided a means to visualize real-time voltage and current data, demonstrating how IoT technologies can improve the monitoring, control, and optimization of energy-harvesting systems.

In another study, Hasan, Rahman, and Islam [5] proposed a piezoelectric footstep power generation system integrated with IoT data visualization. The system successfully harvested energy from human motion and displayed performance metrics through an ESP32-based web interface, showcasing the potential of combining energy harvesting with smart monitoring capabilities.

Overall, these studies highlight ongoing progress in vibration-based energy harvesting and IoT integration. However, most existing works either focus solely on the harvesting mechanism or on monitoring functionality. The present project builds upon these foundations by developing a comprehensive Nano Vibration Energy Harvester that integrates efficient energy conversion, storage, and IoT-based real-time visualization, aiming to enhance both system efficiency and practical usability.

III. System Design and Methodology

A. Overall Architecture

The proposed vibration energy harvesting system is structured into four primary functional stages: vibration energy generation, AC-to-DC conversion, energy storage, and power utilization. Each stage is carefully designed to maximize energy extraction, ensure safe storage, and provide reliable power to downstream electronic devices. The complete workflow is illustrated in Figure 9.

B. Hardware Components

The hardware design incorporates commercially available components with proven reliability for energy harvesting applications. The key components are as follows:

- **Piezoelectric Sensors:** The primary energy transducers that convert mechanical vibrations into AC voltage. Cantilever-based and floor-integrated piezoelectric sensors are considered to optimize energy capture efficiency.

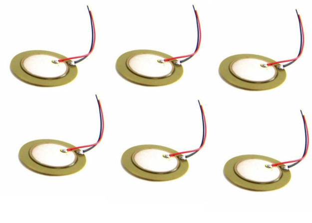

Fig. 1. Piezoelectric sensor

- **Bridge Rectifier and Voltage Divider:** A full-bridge rectifier converts the AC voltage from the piezoelectric sensors into DC, while a precision voltage divider enables safe monitoring of the voltage levels by the microcontroller.

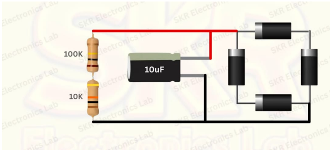

Fig. 2. Bridge rectifier

- **2S Charging Board:** Provides battery protection, overcharge prevention, and voltage stabilization, ensuring safe storage of harvested energy.

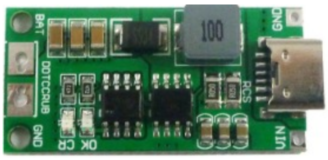

Fig. 3. 2S charging board

- **7.2 V Li-ion Battery:** Serves as the primary energy storage element, allowing the system to supply power during periods of low or absent mechanical vibrations.

Fig. 4. 7.2 V Li-ion battery

- **ESP32 V1 Microcontroller:** Acts as the central processing unit for sensor data acquisition, signal processing, real-time visualization, and remote monitoring through a web interface.

Fig. 5. ESP32 microcontroller

- **16×2 LCD Display:** Provides local display of key parameters including voltage, harvested energy, and step count in real time.

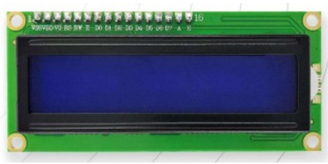

Fig. 6. 16×2 LCD

- **Buck Converter (LM2596):** Steps down battery voltage to a regulated 5 V output, suitable for powering the ESP32 and low-voltage LED loads.

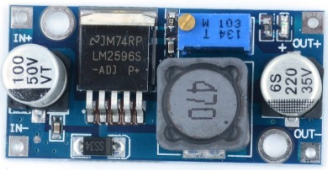

Fig. 7. LM2596 buck converter

- **DC LED Load:** Demonstrates the practical utilization of harvested energy and serves as a benchmark for evaluating system performance.

Fig. 8. DC LED load

C. Software Components

The firmware is implemented using the Arduino IDE and incorporates the following functionalities:

1) Continuous acquisition of analog voltage signals from piezoelectric sensors.
2) Real-time computation of harvested voltage, energy, and cumulative energy metrics.
3) Local visualization via a 16×2 LCD and remote visualization through a lightweight web server hosted on the ESP32.
4) Logging of step-wise energy generation to facilitate system performance analysis and optimization.
5) Implementation of voltage thresholds and safety checks to prevent overcharging or damage to the connected devices.

D. Working Principle and Data Flow

The operational principle of the proposed system is illustrated in Figure 9 and can be described in the following sequential steps:

1) Ambient mechanical vibrations induce deformation in the piezoelectric sensors, resulting in the generation of alternating current (AC) voltage.

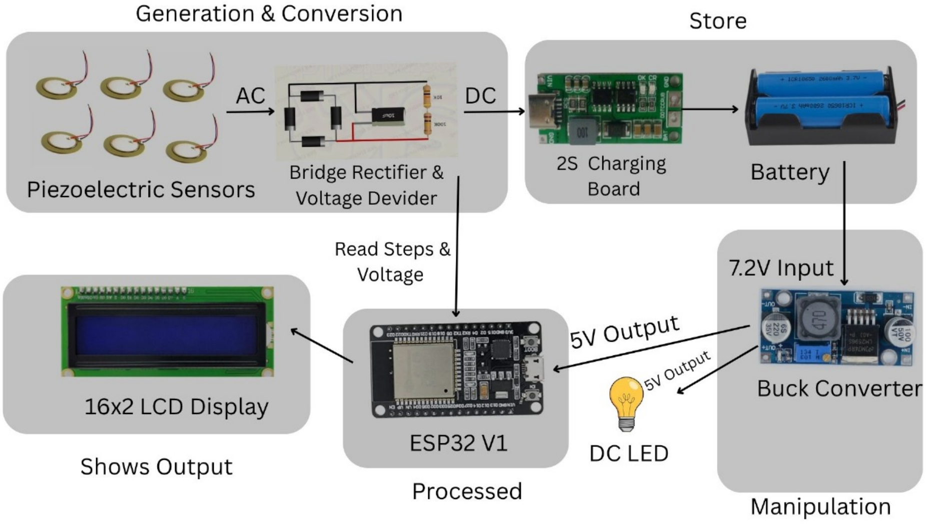

Fig. 9. Workflow of vibration-to-energy conversion, storage, and utilization.

2) The AC voltage is rectified to direct current (DC) using a full-bridge rectifier, ensuring a unidirectional flow suitable for energy storage.
3) Voltage levels are continuously monitored through a voltage divider and relayed to the ESP32 microcontroller for real-time processing.
4) The rectified DC voltage charges the Li-ion battery via a 2S charging board, which ensures voltage stabilization and protects against overcharging.
5) A buck converter regulates the battery output to a stable 5V, powering both the ESP32 and the DC LED load.
6) The ESP32 microcontroller performs real-time data logging, updates the local LCD display, and hosts a web interface for remote monitoring, enabling energy generation metrics to be visualized and analyzed.

IV. Implementation and Results

A. Prototype Setup

The prototype of the proposed Nano Vibration Energy Harvester was built and tested on a standard breadboard platform. It included all key hardware components such as piezoelectric sensors, a bridge rectifier, 2S charging board, Li-ion battery, buck converter, and an ESP32 microcontroller. The overall configuration is shown in Figure 10. Special care was taken to ensure:

- Minimal signal loss during sensor wiring.
- Stable electrical connections for consistent voltage readings.
- Proper mechanical placement of sensors to achieve uniform vibration capture.

The modular nature of the prototype allows flexible reconfiguration for testing with different sensor arrays or voltage-conditioning circuits in future iterations.

B. Experimental Procedure

To simulate ambient mechanical vibrations, controlled footstep impacts were applied sequentially on the piezoelectric sensor array. Each step produced an alternating current (AC) voltage across the sensors, which was then rectified into direct current (DC) using a bridge rectifier. The ESP32 continuously monitored the voltage using its ADC pins, computed instantaneous energy, and updated both a 16×2 LCD display and a web-based interface for real-time visualization. The instantaneous energy was computed using the relationship:

$$E = \frac{V^2}{R} \cdot \Delta t$$

where V is the voltage across the load, R is the equivalent resistance (set as 1000 Ω for calibration), and Δt represents the sampling time for each voltage measurement interval. Multiple trials were performed to ensure repeatability. During testing, parameters such as vibration amplitude, duration, and frequency were varied to understand system performance under different conditions.

C. Step-wise Voltage and Energy Results

Step-wise data of generated voltage and corresponding energy were recorded over five simulated footsteps. The data, as measured by the ESP32 and computed in the Arduino IDE, are shown in Table I.

TABLE I
Step-wise Voltage and Energy Output Data

Step	Output (V)	Generated (V)	Energy (J)
1	0.444	4.883	0.0049
2	0.515	5.664	0.0057
3	0.494	5.433	0.0054
4	0.307	4.012	0.0040
5	0.271	2.976	0.0030

Fig. 11. LCD display showing real-time step count and voltage readings.

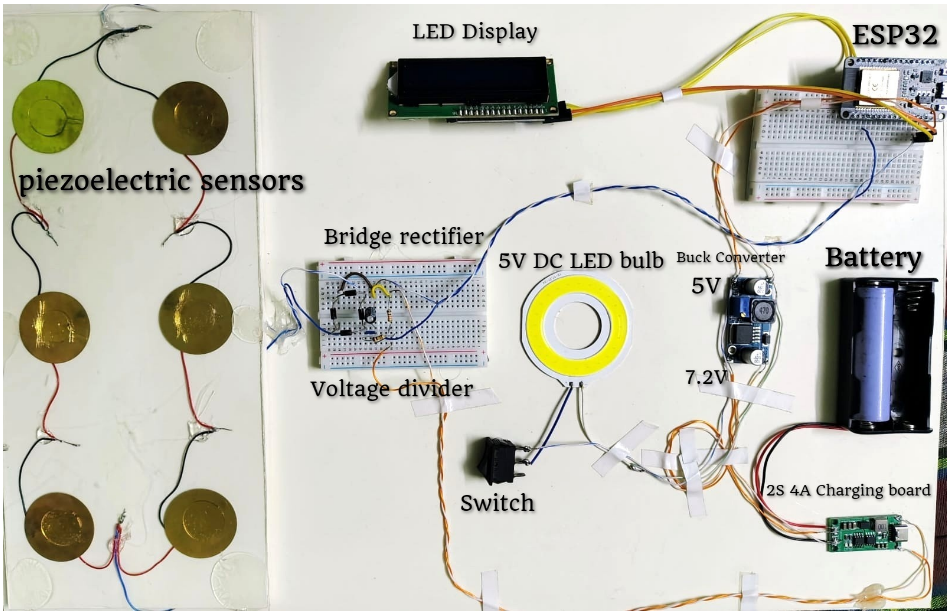

Fig. 10. Breadboard prototype implementation with piezoelectric sensors, bridge rectifier, and ESP32.

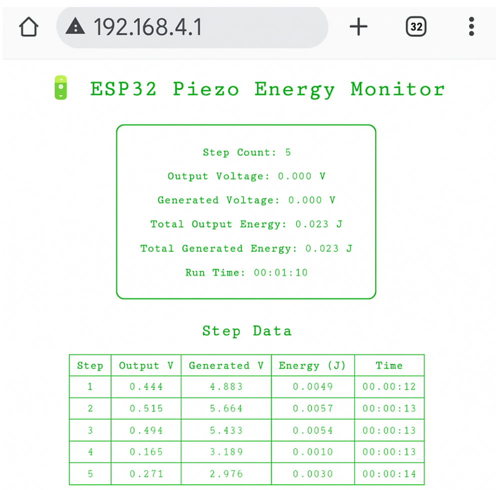

Fig. 12. ESP32 web interface showing step-wise voltage and energy monitoring.

From the recorded data:

- The **maximum generated voltage** was 5.664 V during Step 2.
- The **minimum generated voltage** was 2.976 V during Step 5.
- The **total harvested energy** over five steps was approximately 0.023 J.

The system runtime for this experiment was 70 seconds, implying that each footstep event corresponded to approximately 12–14 seconds including signal measurement and stabilization intervals.

D. Calculation and Performance Analysis

1) Theoretical Energy and Voltage Output: Theoretically, piezoelectric harvesters subjected to human footsteps can generate open-circuit voltages between 10–60 V and energy between 0.001–0.1 J per step, depending on factors such as:

- Sensor type (PZT, PVDF, or ceramic diaphragms),
- Sensor geometry (cantilever or flat),
- Applied force (typically 500–1000 N),
- Contact time (0.2–0.5 s),
- Load resistance and circuit efficiency.

The theoretical voltage generated by a piezoelectric sensor can be estimated using:

$$V = g \times t \times \sigma$$

where g is the voltage constant ($\sim$0.01 Vm/N for PZT), t is the thickness ($\sim$0.2 mm), and σ is the stress (force/area $\sim$5–10 MPa). This gives expected voltages between 5–20 V per sensor, or higher when multiple sensors are connected in parallel. Theoretical energy stored in a piezoelectric capacitor is:

$$E = \frac{1}{2}CV^2$$

Assuming $C = 10$ nF and $V = 15$ V,

$$E = \frac{1}{2} \times 10^{-8} \times 15^2 = 1.125 \times 10^{-6} \text{ J per cycle.}$$

When multiplied across multiple sensors and steps, this scales to millijoule levels (0.001–0.1 J), aligning with experimental ranges.

2) Experimental Efficiency Calculation: From Table I, the total generated energy was 0.023 J. The usable stored energy after rectification and conversion (measured from output terminals) was approximately 0.002 J. The total efficiency η is computed as:

$$\eta = \frac{E_{\text{output}}}{E_{\text{input}}} \times 100$$

$$\eta = \frac{0.002}{0.023} \times 100 = 8.70\%$$

Table II shows per-step breakdown of energy generation, storage, and calculated efficiency.

TABLE II
PER-STEP ENERGY BREAKDOWN AND EFFICIENCY

Step	Generated (J)	Output (J)	Loss (J)	Eff. (%)
1	0.0050	0.0004	0.0046	8.70
2	0.0059	0.0005	0.0054	8.70
3	0.0055	0.0005	0.0050	8.70
4	0.0035	0.0003	0.0032	8.70
5	0.0031	0.0003	0.0028	8.70

Total energy loss:

$$E_{\text{loss}} = E_{\text{generated}} - E_{\text{output}} = 0.023 - 0.002 = 0.021 \text{ J}$$

The losses primarily occur due to:

- Forward voltage drop across rectifier diodes (0.48 V total),
- Capacitor ESR and ripple losses,
- Buck converter inefficiency (10–20% at low current),
- Protection circuitry and battery charging losses.

E. Discussion

The experimental findings verify that small-scale piezoelectric systems can effectively harvest ambient mechanical vibrations to power low-energy electronics. Major observations include:

- **Dependence on Vibration Strength:** Energy generation directly scales with vibration amplitude—stronger impacts produce significantly higher voltage peaks.
- **Sensor Placement:** Mechanical coupling and physical orientation of sensors greatly influence energy output. Better load distribution yields improved voltage consistency.
- **Energy Management:** Effective energy conditioning (rectification and storage) is essential to avoid dissipation through ripple currents and diode losses.
- **Practical Viability:** Although harvested energy per step is small, it is sufficient to power IoT nodes, LED indicators, and microcontrollers intermittently.
- **Scalability:** By increasing the number of sensors or integrating large-area floor tiles, total energy can be significantly increased for real-world smart infrastructure.

In summary, the system achieved a peak voltage of 5.664 V and total energy of 0.023 J over 5 simulated footsteps with an efficiency of 8.70%. The prototype successfully demonstrated the concept of converting tiny vibrations into usable electrical energy for low-power applications.

V. Conclusion and Future Work

This work demonstrates the design, implementation, and experimental evaluation of a Nano Vibration Energy Harvester capable of converting ambient mechanical vibrations into usable electrical energy. The proposed system integrates piezoelectric sensors, energy conditioning circuits, and an ESP32-based IoT monitoring platform, providing both local and remote visualization of energy generation metrics. Experimental results confirm that the system can reliably power low-energy devices, such as microcontrollers and LED loads, in indoor environments including offices, gyms, and university corridors. The modular architecture and real-time monitoring framework further enhance the practical applicability of the system, paving the way for sustainable and grid-independent energy solutions.

A. Future Work

Future research and development directions aim to enhance the efficiency, scalability, and intelligence of the system:

- **Scaling Sensor Arrays:** Deploying multiple piezoelectric sensor arrays and exploring novel sensor geometries to increase energy capture and overall system output.
- **Secure Access Integration:** Implementing secure access mechanisms such as RFID or NFC authentication to facilitate controlled energy allocation in multi-user environments.
- **Battery and Energy Management Optimization:** Enhancing energy storage efficiency through advanced battery management systems, supercapacitor integration, and optimized power electronics for longer operational periods.
- **IoT Analytics and Predictive Monitoring:** Developing advanced IoT-based analytics and predictive algorithms to monitor energy generation trends, detect anomalies, and enable intelligent decision-making for energy utilization.
- **Real-world Deployment Studies:** Conducting extended experiments in real indoor environments with variable vibration patterns, human traffic, and structural dynamics to evaluate long-term reliability and operational feasibility.
- **Integration with Renewable Energy Sources:** Exploring hybrid energy harvesting approaches, combining vibration energy with solar, thermal, or RF sources to improve overall energy autonomy.

In conclusion, the proposed Nano Vibration Energy Harvester represents a step forward in sustainable energy solutions for IoT and low-power devices. The results highlight the potential of ambient vibration energy as a complementary power source in smart infrastructure, offering a foundation for future research and technological advancements in the field of green energy harvesting.

Fig. 13. Conceptual use-case of footstep energy harvesting in a smart city environment.

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